MCMC in Mixture Models

Amartya Kumar Maulik [*], Sayan Das [†], Soham Ghosh[‡]

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Department of Mathematics & Statistics, Indian Institute of Technology, Kanpur

Abstract

The finite mixture model provides a natural representation of heterogeneity in a finite number of latent classes. It concerns modelling a statistical distribution by a mixture (or weighted sum) of other distributions. However, our study sheds some light on infinite mixture models, in order to model data drawn from an unknown mixture distribution with unknown number of clusters. In this article, We have used the infinite Dirichlet mixture model which is famous in the literature for clustering and density estimation problems. In our approach, the determination of the number of clusters is sidestepped by assuming an infinite number of clusters. The effectiveness of our approach is tested on both simulated data and a real life data.

1 Introduction

Mixture distributions arise in practical problems when the measurements of a random variable are taken under two different conditions. For instance, (Gelman & Stern, 2014) presents the distribution of heights in a population of adults. This distribution reflects the mixture of males and females in the population. For the greatest flexibility, consistent with the prevailing general hierarchical modelling strategies, these distributions are constructed as mixtures of simpler forms. To elucidate on this, it is best to model male and female heights as separate univariate, perhaps normal distributions, rather than a single bimodal distribution. Thus mixture models can be used in problems where the population of sampling units consists of a number of sub-populations within each of which a relatively simple model applies.

Suppose that, based on substantive considerations, it is considered desirable to model the distribution of $\boldsymbol{y} = (y_1, \ldots, y_n)$ as a mixture of H components. It is assumed that it is not known which mixture component underlies each particular observation. When H is finite, for $h = 1, 2, \ldots, H$, the h−th component distribution, $f_h(y_i|\theta_h)$, is assumed to depend on a parameter vector $\boldsymbol{\theta}_h$. The parameter denoting the proportion of the population from component h is π_h, with

$$\sum_{h=1}^{H} \pi_h = 1$$

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It is common to assume that the mixture components are all from the same parametric family, such as the Normal, with different parameter vectors. The sampling distribution of $\boldsymbol{y}$ in that case is

$$p(y_i|\boldsymbol{\theta}, \boldsymbol{\pi}) = \pi_1 f(y_i|\theta_1) + \pi_2 f(y_i|\theta_2) + \ldots + \pi_H f(y_i|\theta_H)$$

Finite mixture models made their first recorded appearance in the modern statistical literature in the nineteenth century in a paper by (Newcomb, 1886). We refer to (McLachlan & Peel, 2000) for a comprehensive survey on the history and applications of finite mixture models. (Pearson, 1894) used a mixture of two univariate Gaussian distributions to analyze a dataset containing ratios of forehead to body lengths for 1,000 crabs, using the method of moments (MOM) to estimate the parameters in the model. In multivariate multi-component settings however, this is rarely practical. Fortunately however, maximum likelihood (ML) estimation is possible when implemented via the **Expectation-Maximization** (E.M.) algorithm and is the method of choice in estimation in finite mixture models.

However, in this report we fixate our attention to infinite mixture models. In many applications it is desirable to allow the model to adjust its complexity to the amount the data. The easiest example to visualize in this case is the task of assigning objects into clusters or groups. This task often involves the specification of the number of groups. However, often times it is not known beforehand how many groups exist. Moreover, in some applications, e.g. modelling topics in text documents or grouping species, the number of examples per group is heavy tailed. This makes it impossible to pre-define the number of groups and requiring the model to form new groups when data points from previously unseen groups are observed. A natural approach for such applications is the use of **Bayesian non-parametric models**. In Bayesian nonparametrics, the number of parameters is itself considered to be a random variable. A prior is defined over an infinite dimensional model space, and inference is done to select the number of parameters. Such models have infinite capacity, in that they include an infinite number of parameters a priori; however, given finite data, only a finite set of these parameters will be used. Unused parameters will be integrated out.

For our purpose, we have used the Dirichlet process, which is an infinite-dimensional generalization of the Dirichlet distribution, to set a prior on unknown distributions. This specific choice of prior is due to the many good properties possessed by the Dirichlet distribution, one such property being conjugacy with the multinomial distribution. This idea is quite popular in contemporary Bayesian literature, as presented by (Hannah & Powell, 2011) in mixtures of Generalized Linear Models, (Gorur & Rasmussen, 2010) in Gaussian Mixture Models. Our report focuses on introducing the idea and motivation of Dirichlet processes in Section 2, whereas in Section 3 , we try to use the Dirichlet process mixtures in order to estimate an unknown mixture density function with unknown number of clusters. The parameters of the model are updated iteratively using different variants of Gibbs sampling procedure namely : **Collapsed Gibbs, Blocked Gibbs** and **Sliced Gibbs** respectively. Section 4 contains some simulation results obtained by implementing a collapsed Gibbs and a blocked Gibbs sampler for a Gaussian mixture model and then for a Poisson mixture model. In Section 5, we have implemented blocked Gibbs sampler on a real life dataset. The advantages and drawbacks of using each sampler are outlined as well in Section 6.

2 Background and motivation

The Dirichlet Distribution

We start by defining the Dirichlet distribution. It is a distribution over the $(K-1)$-dimensional simplex; that is, it is a distribution over the relative values of K components, whose sum is restricted to be 1. It is parameterized by a $K-$ dimensional vector $(\alpha_1, \alpha_2, \ldots, \alpha_K)$, where $\alpha_k \geq 0$ for all k and $\sum_k \alpha_k > 0$. Now if a vector $\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots, \pi_K)$ is said to follow Dirichlet$(\alpha_1, \ldots, \alpha_K)$, then $\pi_k \geq 0$ for all k and $\sum_{k=1}^{K} \pi_j = 1$; the distribution is given by :

$$f(\boldsymbol{\pi}, \alpha_1, \ldots, \alpha_K) = \frac{\prod_{k=1}^{K} \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^{K} \alpha_k)} \prod_{k=1}^{K} \pi_k^{\alpha_k - 1}.$$

The expectation of the distribution is :

$$E[(\pi_1, \ldots, \pi_K)] = \left(\frac{\alpha_1}{\sum_k \alpha_k}, \ldots, \frac{\alpha_K}{\sum_k \alpha_k} \right).$$

It is also a well known fact that the Dirichlet distribution is the conjugate of the Multinomial distribution. That is, if $\boldsymbol{\pi} \sim$ Dirichlet$(\alpha_1, \ldots, \alpha_K)$ and $\boldsymbol{x} = (x_1, \ldots, x_n) \sim$ Multinomial$(\boldsymbol{\pi})$, then

$$\begin{aligned} p(\boldsymbol{\pi} | \boldsymbol{x}) &\propto \frac{\prod_{k=1}^{K} \Gamma(\alpha_k + m_k)}{\Gamma \sum_{k=1}^{K} (\alpha_k + m_k)} \prod_{k=1}^{K} \pi_k^{\alpha_k + m_k - 1} \\ &= \text{Dirichlet}(\alpha_1 + m_1, \ldots, \alpha_K + m_K). \end{aligned}$$

We now present a way of defining a Dirichlet distribution having an infinite number of parameters (Xing, 2014).

Going from finite to infinite dimensions

Consider the following scheme :

- Begin with a two-component Dirichlet distribution with scaling parameter α divided equally between both components,

$$\pi^{(2)} = (\pi_1^{(2)}, \pi_2^{(2)}) \sim \text{Dir}\left(\frac{\alpha}{2}, \frac{\alpha}{2}\right).$$

- Split off components according to the expansion rule :

$$\theta_1^{(2)}, \theta_2^{(2)} \sim \text{Beta}\left(\frac{1}{2} \times \frac{\alpha}{2}, \frac{1}{2} \times \frac{\alpha}{2}\right).$$

  We obtain $\pi^{(4)} = (\theta_1^{(2)} \pi_1^{(2)}, (1 - \theta_1^{(2)}) \pi_1^{(2)}, \theta_2^{(2)} \pi_2^{(2)}, (1 - \theta_2^{(2)}) \pi_2^{(2)})$.

- Repeat this process such that

$$\pi^{(k)} \sim \text{Dir}\left(\frac{\alpha}{k}, \frac{\alpha}{k}, \ldots., \frac{\alpha}{k}\right).$$

As $k \to \infty$, we will get a Dirichlet distribution over the infinite dimensional space.

The Dirichlet Process

The previous construction motivates us to define the Dirichlet Process. Let the base P_0 be a distribution over some space Ω. Let

$$\boldsymbol{\pi} \sim \lim_{K\to\infty} \text{Dirichlet}\left(\frac{\alpha}{K}, \dots, \frac{\alpha}{K}\right).$$

For each point in this Dirichlet distribution, we associate a draw from the base measure :

$$\theta_k \sim P_0 \text{ for } k = 1, 2, \dots$$

Then

$$P := \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k}$$

is an infinite discrete distribution over the continuous space Ω. Here $\delta(.)$ is the Dirac measure. We write this as a Dirichlet Process : $P \sim \text{DP}(\alpha, P_0)$. More rigorously, we can define a Dirichlet Process as follows.

Suppose the sample space Ω is partitioned into measurable subsets (bins) $B_1, \dots, B_k$. Let P be a random probability measure over $(\Omega, \mathcal{B})$ where $\mathcal{B}$ is the collection of all possible subsets of Ω. The probability measure will assign probabilities to these subsets given by :

$$P(B_1), \dots, P(B_k) = \int_{B_1} f(y)dy, \dots, \int_{B_k} f(y)dy.$$

Now, if P is a random probability measure, then these bin probabilities are random variables. So, we can take a simple conjugate prior for the bin probabilities corresponding to the Dirichlet distribution. We could let

$$P(B_1), \dots, P(B_k) \sim \text{Dirichlet}(\alpha P_0(B_1), \dots, \alpha P_0(B_k)),$$

where P_0 is a base probability measure providing an initial choice of P such as a Gaussian distribution, and α is a prior concentration parameter which controls the degree of shrinkage of P towards P_0. This definition of the Dirichlet Process and properties of the Dirichlet distribution imply

$$P(B) \sim \text{Beta}(\alpha P_0(B), \alpha(1 - P_0(B))), \ \forall \ B \in \mathcal{B},$$

that the marginal random probability assigned to any subset B of the support is Beta distributed. Also, the prior mean has the form

$$E(P(B)) = P_0(B), \ \forall \ B \in \mathcal{B}.$$

This indicates that the prior for P is centered on P_0, Additionally, the prior variance is

$$V(P(B)) = \frac{P_0(B)(1 - P_0(B))}{1 + \alpha}, \ \forall \ B \in \mathcal{B},$$

clearly implying the fact that α is a precision parameter controlling the variance.

The DP prior inherits the conjugacy property from a Dirichlet distribution which makes inference straightforward. Let $y_i \overset{\text{i.i.d}}{\sim} P$, for $i = 1, \ldots, n$ and $P \sim \text{DP}(\alpha, P_0)$. Then for any measurable partition $B_1, \ldots, B_k$, we have

$$P(B_1), P(B_2), \ldots, P(B_k)|\boldsymbol{y} \sim \text{Dirichlet}\left(\alpha P_0(B_1) + \sum_{i=1}^{n} \mathbb{I}_{y_i \in B_1}, \ldots, \alpha P_0(B_k) + \sum_{i=1}^{n} \mathbb{I}_{y_i \in B_k}\right).$$

From this, it is easy to see that :

$$P|\boldsymbol{y} \sim \text{DP}(\alpha P_0 + \sum_i \delta_{y_i}).$$

The Stick Breaking Construction

The stick-breaking representation is a direct constructive representation of the Dirichlet process. It helps us sample from a Dirichlet Process. This representation allows us to induce $P \sim \text{DP}(\alpha P_0)$ by letting

$$P(.) = \sum_{h=1}^{\infty} \pi_h \delta_{\theta_h}(.), \quad \pi_h = V_h \prod_{l<h}(1 - V_l), \quad V_h \sim \text{Beta}(1, \alpha), \quad \theta_h \sim P_0,$$

where δ_θ denotes a degenerate distribution with all its mass at θ, the **atoms** $(\theta_h)_{h=1}^{\infty}$ are generated independently from the base distribution P_0, π_h is the probability mass at θ_h, and these probability masses are generated from this process that guarantees that the weights sum to 1. The construction is elaborated below :

Consider a stick of unit length representing the total probability to be allocated to all the atoms. We draw V_1 from a Beta$(1, \alpha)$ distribution randomly and allocate $\pi_1 = V_1$ probability weight to the randomly generated first atom $\theta_1 \sim P_0$. Iteratively sample a random observation V_h , $h = 2, 3, \ldots$ from Beta$(1, \alpha)$ and break off a fraction of V_h of the remaining portion of the stick. This is the weight of the h^{th} atom for $h = 2, 3 \ldots$. Sample a location for this atom from the base measure P_0 and repeat.

This will give an infinite number of atoms, but as the process continues, the weights of the atoms will decrease to be negligible. This motivates truncation approximations, in which the full Dirichlet Process is approximated using a finite number of clusters.

3 Density estimation in an infinite mixture model

Now, we consider the problem of estimating a general kernel mixture model for the density

$$f(y|P) = \int \mathcal{K}(y|\theta) dP(\theta), \tag{1}$$

where $\mathcal{K}(.|\theta)$ is a kernel with parameter θ and P is a mixture measure. In case of a infinite kernel mixture we choose a prior distribution for P. If this prior is chosen to be a DP prior, then it becomes a DP mixture model. A DP prior on P is given as

$$f(y) = \sum_{k=1}^{\infty} \pi_k \mathcal{K}(y|\theta_k^*), \tag{2}$$

where $\pi \sim stick(\alpha)$ i.e. the probability weights are sampled from a DP stick breaking process with precision parameter α. Here, $\theta_k \overset{ind}{\sim} P_0$ for $k = 1, 2, \ldots \infty$. Although the number of mixture component is set to infinite, it does not imply that infinite number of components are occupied by the subjects in the sample, rather it allows flexibility by introducing new mixture components as subjects are added. Now, consider sampling from y_i from the above mentioned infinite mixture model using the hierarchical conditions,

$$y_i \sim \mathcal{K}(\theta_i), \quad \theta_i \sim P, \quad P \sim DP(\alpha P_0),$$

Marginalizing out P will induce the prior distribution on the subject specific parameter $\boldsymbol{\theta} = (\theta_1, \theta_1, \ldots, \theta_n)$, known as Polya urn predictive rule,

$$p(\theta_i | \theta_1, \theta_2, \ldots, \theta_{i-1}) \sim \frac{\alpha}{\alpha + i - 1} P_0(\theta_i) + \sum_{j=1}^{i-1} \frac{1}{\alpha + i - 1} \delta_{\theta_j}, \tag{3}$$

which is a mixture of base probability measure P_0 and previous subject's parameter values.

The Chinese Restaurant metaphor

It is useful metaphor for helping to understand the Dirichlet Process.

Consider a restaurant with infinitely many tables. The first customer sits at a table with dish θ_1^*. The second customer sits at the first table with probability $\frac{\alpha}{1+\alpha}$ or a new table with probability $\frac{1}{1+\alpha}$. This process continues with the i-th customer sitting at an occupied table with probability proportional to the number of previous customers at that table and sitting at a new table with probability proportional to α. Each occupied table in the (infinite) restaurant represents a different cluster of subjects, with new clusters being added. The number of clusters depends probabilistically on the number of subjects n with new clusters introduced as needed as additional subjects are added to the sample.

Using exchangeability of the subjects $i = 1, \ldots, n$, one can obtain the conditional prior distribution of θ_i given $\boldsymbol{\theta_{(-i)}} = (\theta_1, \theta_2, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_n)$ as,

$$p(\theta_i | \boldsymbol{\theta_{(-i)}}) = \frac{\alpha}{\alpha + n - 1} P_0(\theta_i) + \sum_{k=1}^{c^{(-i)}} \frac{n_k^{(-i)}}{\alpha + n - 1} \delta_{\theta_k^{*(-i)}}, \tag{4}$$

where $\theta_k^*, k = 1, 2, \ldots, c^{(-i)}$ are unique values of $\boldsymbol{\theta_{(-i)}}$ and $n_k^{(-i)} = \sum_{j \neq i} \mathbb{I}(\theta_j = \theta_k^*)$.

If P_0 is conjugate to the kernel then updating the full conditional prior with the data will result in a conditional posterior distribution of the same form with updated weights on the components and updated P_0. In particular, when the kernel is a normal kernel with unknown mean and variance and P_0 is a normal-gamma prior distribution, it induces similar form of the conditional posterior. Potentially, one can update the θ_i's one at a time from these full conditional posterior distributions in implementing Gibbs sampling. However, this approach tends to have **poor mixing**.

Instead, we implement a collapsed Gibbs sampler which updates the allocation of subjects to clusters and the cluster-specific parameters proceeding as follows :

Collapsed Gibbs Sampler

A marginal Gibbs sampler separately updates the allocation of subjects to clusters and the cluster-specific parameters , proceeds as follows . Let $\boldsymbol{\theta}^* = (\theta_1^*, \theta_2^*, .., \theta_k^*)$ denote the unique values of $\boldsymbol{\theta}$ and let $S_i = c$ if $\theta_i = \theta_c^*$ so that S_i denotes allocation of subject i to a cluster.

Algorithm:

1. Update the allocation S by sampling from the multinomial conditional posterior with
$$\Pr(S_i = c \mid -) = \begin{cases} n_c^{(-i)} \mathcal{K}(y_i \mid \theta_c^*) & c = 1, ..., k^{(-i)} \\ \alpha \int \mathcal{K}(y_i \mid \theta) dP_0(\theta) & c = k^{(-i)} + 1. \end{cases}$$
If $S_i = k^{(-i)} + 1$, then subject i is allocated to a singleton cluster.

2. Update the unique values θ^* by sampling from
$$p(\theta_c^* \mid -) \propto P_0(\theta_c^*) \prod_{i:S_i=c} \mathcal{K}(y_i \mid \theta_c^*),$$
which is simply the posterior distribution under the parametric model that assigns prior distribution P_0 to the parameters θ_c^* and then updates this prior with the likelihood for the subjects in cluster c.

Here, $n_c^{(-i)} = \sum_{j \neq i} \mathbb{I}(\theta_j = \theta_c^*)$, $n_c = \sum_{j} \mathbb{I}(\theta_j = \theta_c^*)$ and $k^{(-i)}$ is the number of unique values in θ^* excluding the ith observation.

Blocked Gibbs Sampler

By marginalizing out the random probability measure P, we give up the ability to conduct inferences on P. By having approaches that avoid marginalization, we open the door to generalizations of DPMs for which marginalization is not possible analytically. One approach for avoiding marginalization is to rely on the construction in (2) Since the stick-breaking construction orders the mixture components so that the weights are stochastically decreasing in the index h, for a sufficiently high index N, we will have that $\sum_{N+1}^{\infty} \pi_h \to 0$. Hence, we can obtain an accurate approximation by letting $V_N = 1$ in the stick-breaking process so that $\pi_h = 0$ for $h = N+1, \ldots, \infty$, with N chosen to be sufficiently large. The truncation approximation to the DP leads to a straightforward MCMC algorithm for posterior computation. It is not clear whether this approach leads to more efficient posterior computation then the collapsed sampler, though the exchangeability of the components in the finite Dirichlet approximation conveys some advantages in terms of mixing.

Algorithm:

1. Update $S_i \in \{1, 2, ..N\}$ by multinomial sampling with
$$\Pr(S_i = c \mid -) = \frac{\pi_c \mathcal{K}(y_i \mid \theta_c^*)}{\sum_{c'=1}^{N} \pi_{c'} \mathcal{K}(y_i | \theta_{c'}^*)} \quad , c' = 1, .., N,$$
where $S_i = c$ if $\theta_i = \theta_c^*$ denotes that subject i is allocated to cluster c.

2. Update the stick breaking weight $V_c,\ c = 1, .., N-1$, from $\text{Beta}(1+n_c, \alpha+\sum_{c'=c+1}^{N} n_{c'})$.
3. Update $\theta_c^*, c = 1, ..N$ exactly as in the finite mixture model , with parameters for unoccupied clusters with $n_c = 0$ sampled from the prior P_0.

Here, $n_c = \sum_j \mathbb{I}(S_i = c))$ is the number of samples in the cth cluster.

Slice Sampler

Slice samplers for DP mixture models were introduced in (Walker, 2007). Like blocked Gibbs samplers, slice samplers do not marginalize over P, but use the stick breaking representation of the process. Slice sampling is well-known for its desirable properties including its fast mixing and its natural potential for parallelization. It induces a random truncation to the infinite clusters. Marginalizing over this random truncation , we would recover the full model. For this, we take a random variable u_i for each data point

$$u_i \mid - \sim U(0, \pi_{S_i}).$$

Conditioned on u_i and S_i , π_k can be sampled according to the blocked Gibbs sampler. We sample indicators according to

$$\Pr(S_i = c \mid -) = \mathbb{I}_{\pi_c > u_i} \mathcal{K}(y_i \mid \theta_c^*).$$

The scheme only represents a finite number of components K such that

$$1 - \sum_{k=1}^{K} \pi_k < \min(u_i).$$

4 Simulation Study

In this simulation example we consider a mixture of k Gaussian densities. Define $\phi(x|\mu, \sigma^2)$ as the value of Gaussian density with mean μ and variance σ^2 at the point x. Consider standard normal distribution as the base measure P_0, so that

$$y_i \sim N(\mu_i, 1), \quad \mu_i \sim P,\ i = 1(1)n; \quad P \sim DP(\alpha P_0), \quad P_0 = N(0,1). \tag{5}$$

In this case the algorithm of collapsed Gibbs sampler is given below (Gelman & Stern, 2014)

Algorithm: Collapsed Gibbs for Gaussian Mixtures

1. Update the allocation S by sampling from the multinomial conditional posterior with
$$\mathbb{P}(S_i = c \mid -\ ) = \begin{cases} n_c^{(-i)} \phi(y_i|\mu_c^*, 1) & c = 1, ..., k^{(-i)} \\ \alpha\ \phi(y_i|0, 2) & c = k^{(-i)} + 1. \end{cases}$$
If $S_i = k^{(-i)} + 1$, then subject i is allocated to a singleton cluster.

2. Update the unique values μ^* by sampling from

$$N\left(\frac{n_c\,\mu_c^*}{n_c+1}, \frac{1}{n_c+1}\right).$$

Here, $n_c^{(-i)} = \sum_{j\neq i} \mathbb{I}(\theta_j = \theta_c^*)$, $n_c = \sum_{j} \mathbb{I}(\theta_j = \theta_c^*)$ and $k^{(-i)}$ is the number of unique values in θ^* excluding the ith observation.

The algorithm of blocked Gibbs sampler is as follows (Gelman & Stern, 2014)

Algorithm: Blocked Gibbs for Gaussian Mixtures

1. Update $S_i \in \{1, 2, ..N\}$ by multinomial sampling with

$$\mathbb{P}(S_i = c \mid -) = \frac{\pi_c \phi(y_i|\mu_c^*, 1)}{\sum_{c'=1}^{N} \pi_{c'} \phi(y_i|\mu_{c'}^*, 1)} \quad , c' = 1, .., N,$$

where $S_i = c$ if $\mu_i = \mu_c^*$ denotes that subject i is allocated to cluster c.

2. Update the stick breaking weight $V_c, c = 1, .., N-1$, from

$$\text{Beta}\left(1 + n_c,\ \alpha + \sum_{c'=c+1}^{N} n_{c'}\right).$$

3. Update $\mu_c^*, c = 1, ..N$ from its conditional posterior,

$$N\left(\frac{n_c\,\mu_c^*}{n_c+1}, \frac{1}{n_c+1}\right).$$

Here, $n_c = \sum_{j} \mathbb{I}(S_i = c))$ is the number of samples in cth cluster.

When the samples are from a distribution of counting data, we generally consider a mixture of Poisson distributions. Define $Poi(x|\lambda)$ as the value of probability mass function of Poisson distribution with parameter λ at the point x. Consider gamma distribution as the base measure P_0, so that,

$$y_i \sim Poi(\lambda_i), \quad \lambda_i \sim P,\ i = 1(1)n; \quad P \sim DP(\alpha P_0), \quad P_0 = Gamma(a, b). \tag{6}$$

In this case the algorithm of Blocked Gibbs sampler is given below (Gelman & Stern, 2014)

Algorithm: Blocked Gibbs for Poisson Mixtures

1. Update $S_i \in \{1, 2, ..N\}$ by multinomial sampling with

$$\mathbb{P}(S_i = c \mid -) = \frac{\pi_c\ Poi(y_i|\mu_c^*)}{\sum_{c'=1}^{N} \pi_{c'}\ Poi(y_i|\mu_{c'}^*)} \quad , c' = 1, .., N,$$

where $S_i = c$ if $\mu_i = \mu_c^*$ denotes that subject i is allocated to cluster c.

2. Update the stick breaking weight $V_c, c = 1, .., N - 1$, from
$$\text{Beta}\left(1 + n_c,\ \alpha + \sum_{c'=c+1}^{N} n_{c'}\right).$$

3. Update $\mu_c^*, c = 1, ..N$ from its conditional posterior,
$$Gamma\left(a + \sum_{i:S_i=c} y_i,\ b + n_c\right).$$

Here, $n_c = \sum_j \mathbb{I}(S_i = c))$ is the size of cth cluster.

4.1 Simulation Result I

Consider the problem of density estimation of a Gaussian mixture model, where the components can be easily identified. First we draw 100 samples from a mixture of four Gaussian distributions,

$$f(x) = 0.3\ \phi(x| - 10, 1) + 0.2\ \phi(x| - 3, 1) + 0.2\ \phi(x|3, 1) + 0.1\ \phi(x|10, 1).$$

Estimated density from collapsed Gibbs sampler taking $\alpha = 0.1$ is

$$\hat{f}_C(x) = 0.28\ \phi(x| - 10.42, 1) + 0.17\ \phi(x| - 3.52, 1) + 0.20\ \phi(x|2.94, 1) + 0.35\ \phi(x|9.90, 1).$$

And estimated density from Blocked Gibbs sampler taking $\alpha = 0.1$ and $N = 20$ is

$$\hat{f}_B(x) = 0.28\ \phi(x| - 10.3, 1) + 0.17\ \phi(x| - 3.48, 1) + 0.20\ \phi(x|2.88, 1) + 0.35\ \phi(x|9.87, 1).$$

From Figure 1, it is observed that both collapsed Gibbs sampler and blocked Gibbs

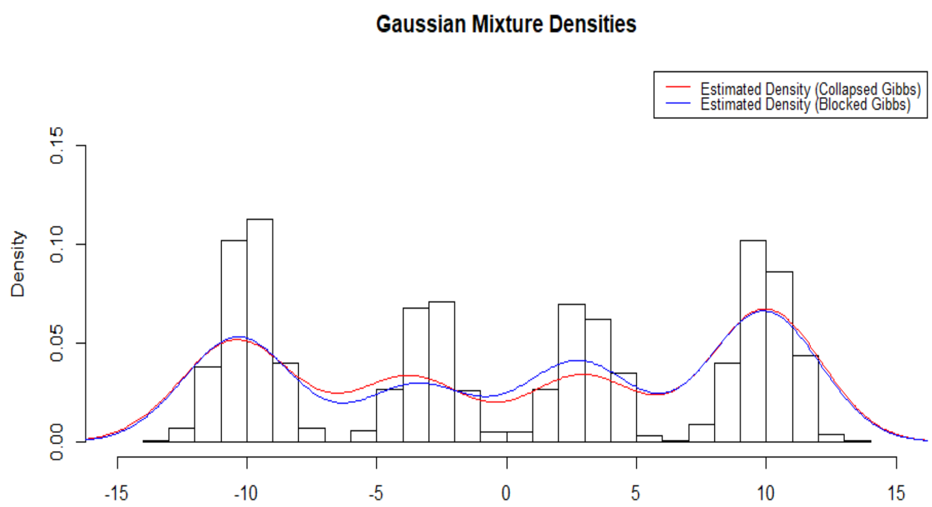

Figure 1: Estimated density plots simulation study I

sampler performed similarly and both of them correctly identified the true density. But blocked Gibbs sampler is much easier to implement than collapsed Gibbs sampler since we are only dealing with a finite number of clusters in blocked Gibbs sampler.

4.2 Simulation Result II

Here we consider the problem of density estimation of a Gaussian mixture model, where the component means are closer to each other and the components cannot be easily identifiable. Draw 100 samples from a mixture of four Gaussian distributions,

$$f(x) = 0.3\ \phi(x|-4,1) + 0.4\ \phi(x|0,1) + 0.2\ \phi(x|4,1) + 0.1\ \phi(x|7,1).$$

Estimated density from collapsed Gibbs sampler taking $\alpha = 0.1$ and $N = 20$ is,

$$\hat{f}_C(x) = 0.31\ \phi(x|-4.14,1) + 0.32\ \phi(x|-0.23,1) + 0.21\ \phi(x|3.70,1) + 0.16\ \phi(x|6.90,1).$$

And estimated density from blocked Gibbs sampler taking $\alpha = 0.1$ and $N = 20$ is,

$$\hat{f}_B(x) = 0.29\ \phi(x|-4.27,1) + 0.35\ \phi(x|-0.17,1) + 0.23\ \phi(x|4.21,1) + 0.13\ \phi(x|6.74,1).$$

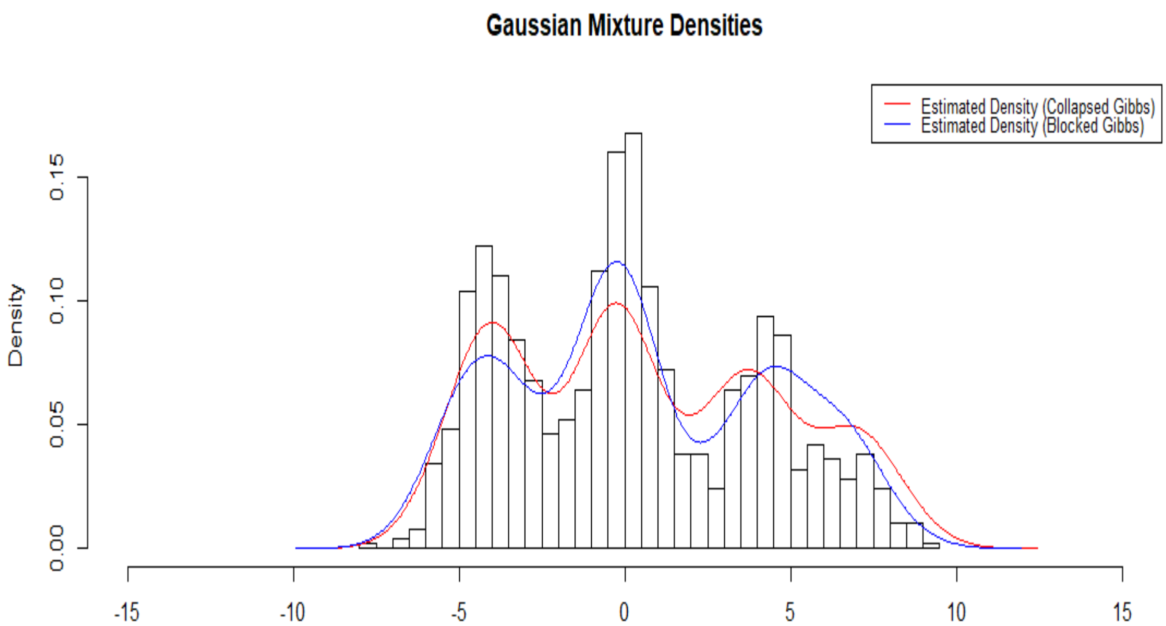

Figure 2: Estimated density plots simulation study II

From Figure 2, it is observed that since a small probability of 0.1 has been assigned to the mixture component $N(7,1)$, the true density seems to exhibit almost 3 clusters, with the final component being dominated by the nearest one. From the values of the estimated means of the four Gaussian components and their mixing probabilities for both the collapsed Gibbs and the blocked Gibbs samplers, it seems that they perform similarly, however the blocked Gibbs sampler is much more efficient with respect to computational time.

4.3 Simulation Result III

Now we shall consider a problem on the distribution of count data. Counts being discrete, it seems natural to use a Poisson kernel with a gamma base measure P_0. Due to computational restraints, we are just using blocked Gibbs Sampler in this case. First we draw 100 samples from a mixture of three Poisson distributions

$$f(x) = 0.3\ Poi(x|2) + 0.4\ Poi(x|15) + 0.3\ Poi(x|30).$$

Estimated density from blocked Gibbs sampler taking $\alpha = 0.1$ and $N = 20$ is,

$$\hat{f}_B(x) = 0.27\ Poi(x|1.7) + 0.42\ Poi(x|14.94) + 0.31\ Poi(x|29.17).$$

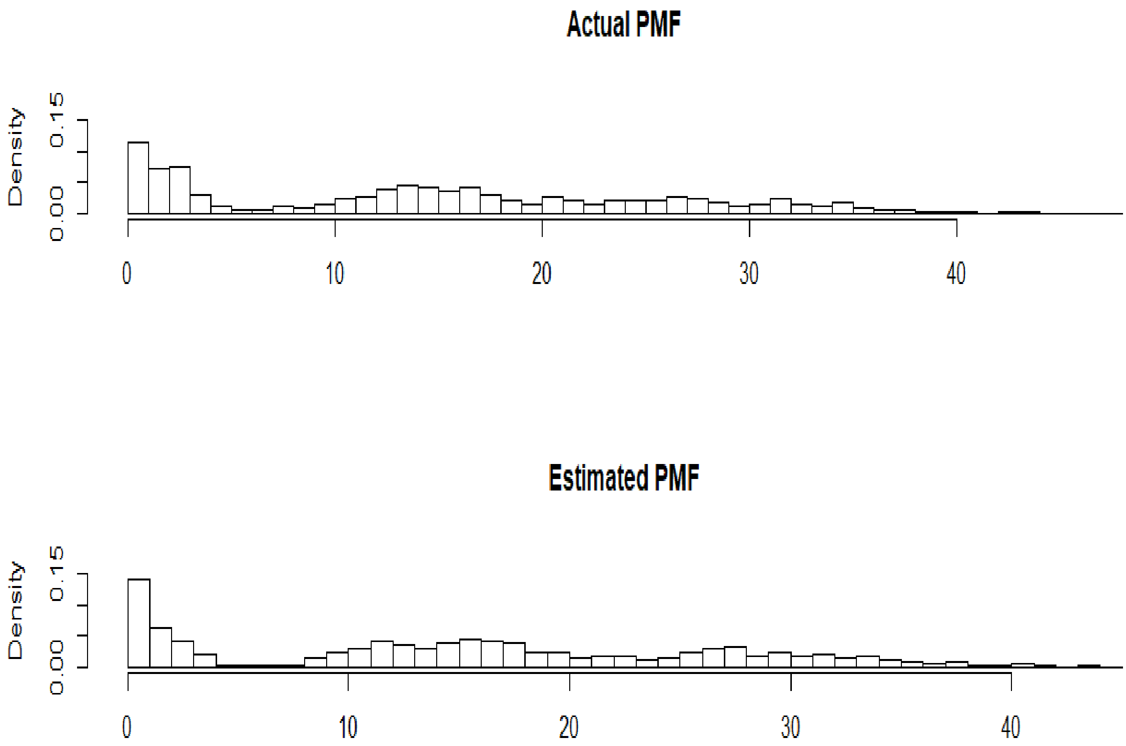

Figure 3: Actual and estimated PMF of Poisson mixtures simulation study III

From both Figure 3 and the model, it is clear that density estimation of the counting data from the Poisson mixture model is quite satisfactory.

5 Application

In this section, we have applied our knowledge on a real life dataset, where we have implemented blocked Gibbs sampler on a mixture Gaussian densities. Define $\phi^*(x|\mu,\tau)$ as the value of Gaussian density with mean μ and precision parameter τ at the point x. This is a reparametrization of $\phi(x|\mu,\sigma^2)$. Consider normal-gamma distribution as the base measure P_0, so that

$$y_i \sim N(\mu_i, \tau_i^{-1}), \quad (\mu_i,\tau_i) \sim P,\ i = 1(1)n, \quad P \sim DP(\alpha P_0), \tag{7}$$

with $P_0(\mu,\tau) = N(\mu|\mu_0, k^{-1}\tau^{-1})Gamma(\tau|a_\tau, b\tau)$. In this case the algorithm of collapsed Gibbs sampler is given below (Gelman & Stern, 2014)

Algorithm: Blocked Gibbs for Two Parameter Gaussian Mixtures

1. Update $S_i \in \{1,2,..N\}$ by multinomial sampling with

$$\mathbb{P}(S_i = c \mid -) = \frac{\pi_c \phi^*(y_i|\mu_c^*, \tau_c^*)}{\sum_{c'=1}^N \pi_{c'} \phi^*(y_i|\mu_{c'}^*, \tau_{c'}^*)} \quad ,c = 1,..,N,$$

   where $S_i = c$ if $\mu_i = \mu_c^*$ denotes that subject i is allocated to cluster c.

2. Update the stick breaking weight $V_c, c = 1,..,N-1$, from

$$\text{Beta}\left(1 + n_c,\ \alpha + \sum_{c'=c+1}^{N} n_{c'}\right).$$

3. Update $\mu_c^*, c = 1,..N$ from its conditional posterior,

$$N(\mu_c^*|\hat{\mu}_c, \hat{k}_c^{-1}\hat{\tau}_c^{-1})\ Gamma(\tau_c|\hat{a}_{\tau_c}, \hat{b}_{\tau_c}).$$

Here $n_c = \sum_j \mathbb{I}(S_i = c)),\ \hat{k} = k + n_c,\ \hat{a}_\tau = a + \frac{n_c}{2},\ \hat{b}_\tau = b + \frac{1}{2}\left(\sum_{i:S_i=c}(y_i - \bar{y})^2 + \frac{n_c k}{\hat{k}}(\bar{y} - \mu_0)^2\right)$ and $\hat{\mu_c} = \frac{k\mu_0 + n_c\bar{y}_c}{\hat{k}}$.

We will now see an application on the `faithful` dataset. Here we are provided with the waiting time between eruptions and the duration of the eruption for the Old Faithful geyser in Yellowstone National Park, Wyoming, USA. Using the Blocked Gibbs Sampler to model the eruptions with a Gaussian mixture model, we get,

$$\hat{f}_{B1}(x) = 0.64\ \phi(x|4.46, 0.22) + 0.36\ \phi(x|1.77, 0.18).$$

We have plotted the eruption times in figure 4. Clearly, it is bimodal with the modes at around 2 seconds and at around 4.5 seconds. Our estimated model seems to fit the dataset very well. The model suggests that about 36 percent of the eruptions are short. Short eruptions have mean 1.77 and variance 0.18 and long eruptions have mean 4.46 and variance 0.22.

Similarly we model the waiting time with a Gaussian mixture model using the Blocked

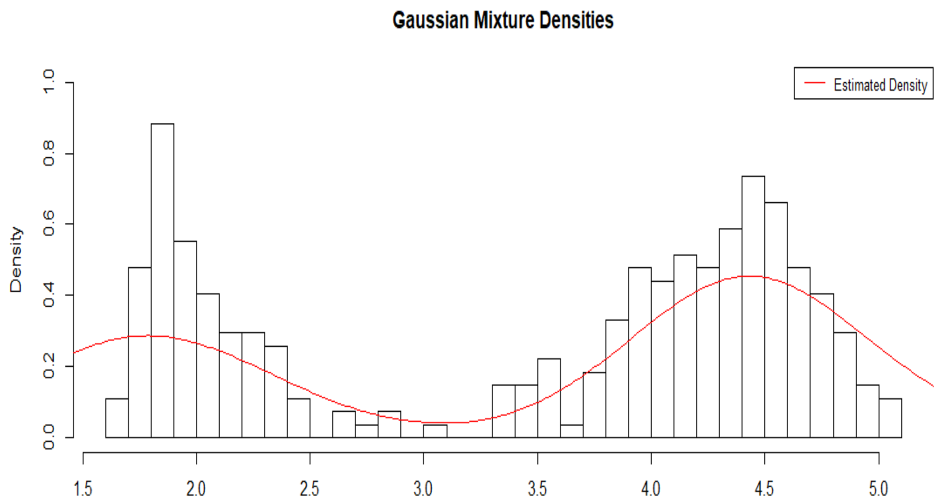

Figure 4: Density plots

Gibbs Sampler and get the following result,

$$\hat{f}_{B_2}(x) = 0.625\ \phi(x|80.09, 33.27) + 0.375\ \phi(x|54.76, 40.62).$$

From the model we can infer that nearly 37.5 percent of the waiting period are short.

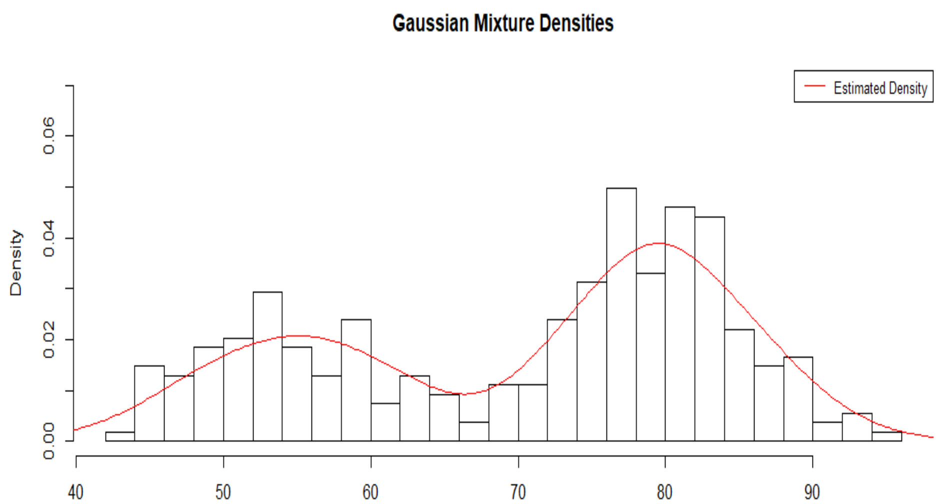

Figure 5: Density plots

Short eruption waiting time have mean 54.76 and variance 40.62 and long eruption waiting time have mean 80.09 and variance 33.27.

6 Conclusions and further developments

Using the Dirichlet Process priors for estimating an unknown mixture density with unknown number of mixture components, we observe the following :

1. It is evident that the conditional posterior probability of allocation to a new cluster is proportional to the DP precision parameter α multiplied by the marginal likelihood for the i^{th} subject's data, obtained in integrating the likelihood $\mathcal{K}(y_i \mid \theta)$ over the prior $\theta \sim P_0$. If α is close to 0 or this marginal likelihood is small relative to the likelihoods for the i^{th} subject's data given allocation to one of the occupied clusters, then the subject i will tend to be allocated to an existing cluster. Hence, both α and P_0 play important roles in controlling the posterior distribution over clustering. As α decreases, there is an increasing tendency to cluster subjects, with a parametric model $y_i \sim \mathcal{K}(\theta)$ for a common θ obtained in the limit as $\alpha \to 0$. In practice, it is common to either set $\alpha = 1$ to favor allocation to few clusters, implying that, in the prior distribution, two randomly selected individuals have a 50% chance of belonging to the same cluster.

2. One can also choose a **gamma hyperprior** for α to allow greater data-adaptivity, with an additional MCMC step included to update α , such as $\alpha \sim \text{Gamma}(a_\alpha, b_\alpha)$, and then updating α during the MCMC analysis. For the blocked Gibbs sampler, the gamma hyperprior is conditionally conjugate with the resulting conditional posterior being

$$\alpha \mid - \sim \textbf{Gamma}(a_\alpha + N - 1, b_\alpha - \sum_{h=1}^{N-1} \log(1 - V_h))$$

3. The more difficult and subtle issue is choice of the base measure P_0. Often the base measure is chosen for computational convenience to be conjugate. However, even in conjugate parametric families such as normal-gamma, we can potentially improve flexibility by placing hyper-parameters on the parameters in P_0. P_0 can be thought of as inducing the prior for the cluster locations. A common approach is to standardize the data $\boldsymbol{y}$ and then choose P_0 to be centered at 0 with close to unit variance. If we set unit variance and do not standardize, then the flatness of P_0 depends on the unit of measurement in the data, that is if we change the unit of the data (say, from inches to miles), we may get completely different results.

4. The Dirichlet Process mixtures do not work particularly well in clustering data which exhibit power law behaviour (Dubey & Williamson, 2014). As a viable alternative, the **Pitman-Yor** process provides an elegant way-out. It is basically a two-parameter extension to the Dirichlet process that allows heavier-tailed distributions over partitions. It also solves the problem of slow mixing when the posterior distribution is multimodal.

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